

Q-CARE Research Evidence Report

A deterministic summary of one completed frozen-model experiment.

EXPERIMENT INFORMATION

Experiment ID	QCARE-20260908-022000-S1NG
Generated	2026-09-08T02:20:00
Input mode	Manual
Feature count	8
Model mode	Compare All

CONFIRMED 8-FEATURE PROFILE

Haemoglobin	14.6
Albumin	0.0
Diabetes mellitus	no
Specific gravity	1.02
Packed cell volume	44.0
Appetite	good
Hypertension	no
Serum creatinine	1.0

MODEL OUTPUTS

VQC	CKD-like
QSVC	non-CKD-like
RBF SVM	non-CKD-like

Experimental decision score (VQC): 0.509. NOT a calibrated disease probability.

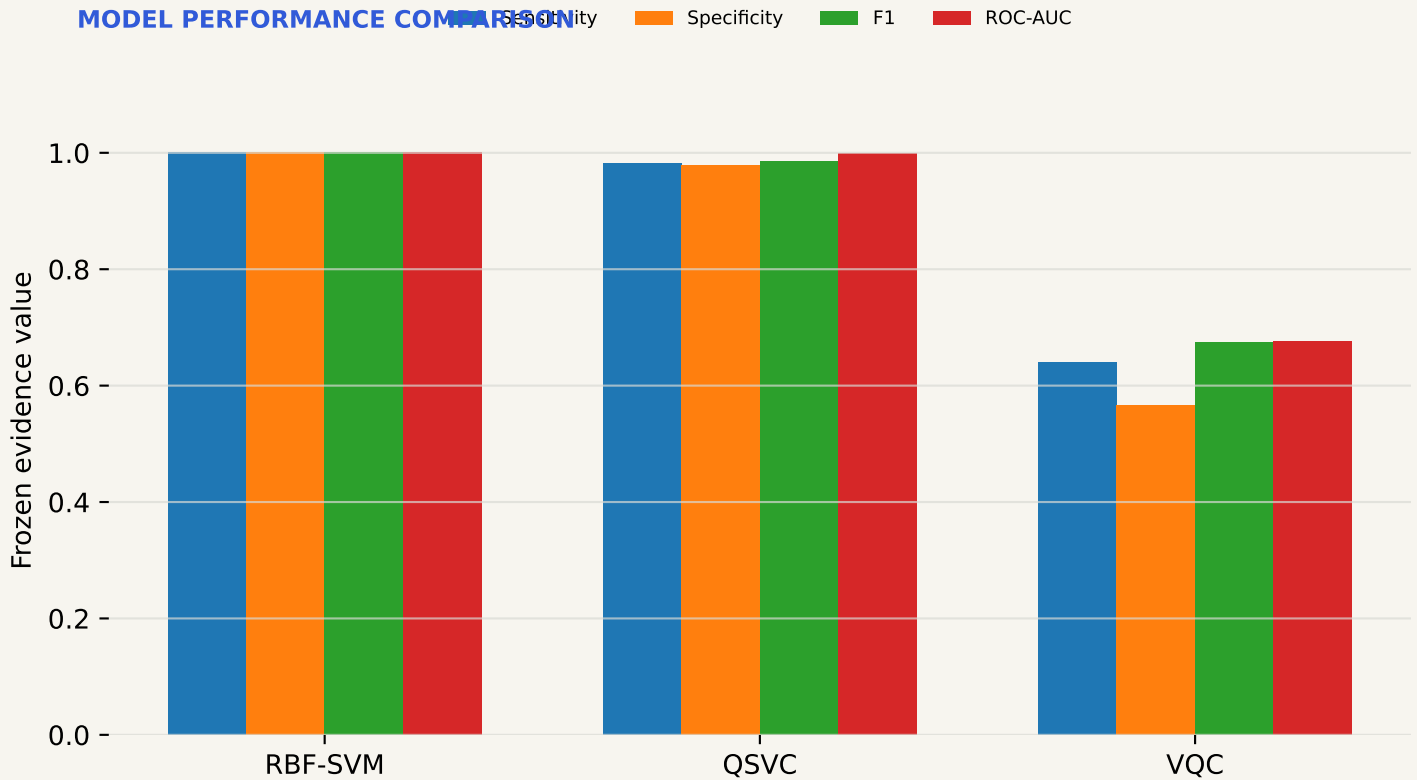
MODEL AGREEMENT

2 OF 3 AGREE. Two of the three research models produced the same class output. Model disagreement should be reviewed alongside model-specific evidence.

Frozen benchmark evidence

Artifact-backed performance and computational cost; no training runs during report generation.

MODEL PERFORMANCE COMPARISON



Values are sourced from the frozen ResultsRepository artifacts. VQC is a separate existing holdout protocol and is not interchangeable with repeated-CV estimates.

COMPUTATIONAL COST

RBF-SVM benchmark/reference runtime 0.000746 s

QSVC frozen benchmark runtime / relative cost 0.346380 s / approximately 464x slower

VQC training runtime 34.57 s

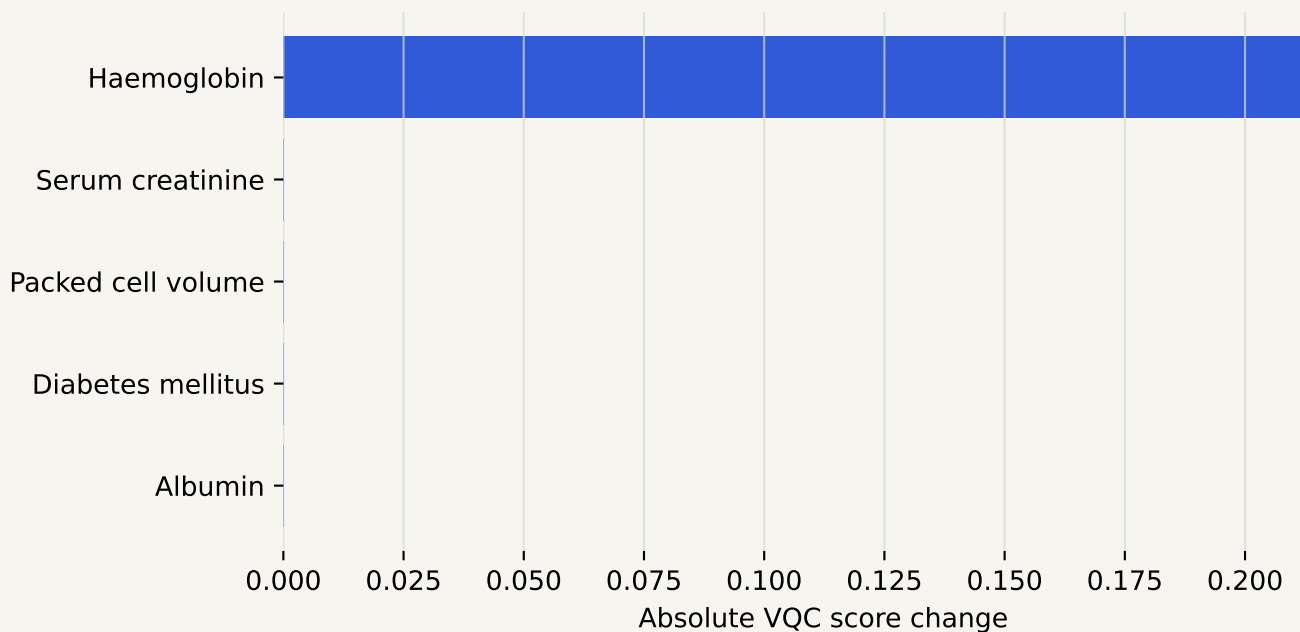
Inference distinction Interactive inference is separate from training cost.

Measured in the project's local exact-statevector benchmarking setup; not a hardware benchmark.

Feature and quantum evidence

Actual stored artifacts and profile-specific evidence only.

FACTORS INFLUENCING THIS MODEL OUTPUT



MODEL INFLUENCE — NOT BIOLOGICAL CAUSATION. These are local perturbation values from the evaluated profile.

FEATURE-BUDGET EVIDENCE

24 original CKD predictors

↓

Leakage-safe feature selection

↓

8-feature primary representation

Selected variables: hemo (Haemoglobin), al (Albumin), dm (Diabetes mellitus), sg (Specific gravity), pcv (Packed cell volume), appet (Appetite), htn (Hypertension), sc (Serum creatinine)

The 8-feature representation retained most of the tested internal predictive performance while using a smaller information budget. Sensitivity: 24 features 0.985, 8 features 0.995; Specificity: 24 features 1.000, 8 features 0.996; F1: 24 features 0.992, 8 features 0.996; ROC-AUC: 24 features 1.000, 8 features 1.000

Quantum, robustness, and transportability

Frozen architecture and Phase 3C evidence boundaries.

QUANTUM MODEL EVIDENCE

QSVC: 8 qubits · ZFeatureMap · reps = 1 · no entanglement · fidelity quantum kernel · C = 0.5.

The quantum circuit creates a similarity representation that is used by the support-vector classifier.

VQC: 8 qubits · ZFeatureMap · RealAmplitudes · linear entanglement · COBYLA · 40 evaluations · 34.57 seconds training. The VQC contains trainable quantum-circuit parameters optimized by a classical optimizer.

VQC status: WEAK — DISPLAY WITH CAUTION. The VQC is retained as a trainable quantum-model research demonstration and was not the strongest predictive model.

ROBUSTNESS & STRESS TESTS

Frozen missingness, numeric perturbation, reduced training size, finite-shot, and simulated-noise summaries are included in the repository artifacts. Missingness, perturbation, and training-size comparisons apply to RBF-SVM and QSVC; finite-shot and simulated-noise findings apply only to tested QSVC settings. No VQC robustness values are invented.

EXTERNAL TRANSPORTABILITY

UCI internal: RBF SVM ROC-AUC 1.000; QSVC ROC-AUC 0.999. BD-KDD stress test: RBF SVM 0.511 (95% CI 0.475-0.546); QSVC 0.525 (95% CI 0.489-0.561). Neither model demonstrated reliable better-than-random discrimination on the BD-KDD cross-cohort stress test. Target comparability: PARTIAL. Internal feature stability did not guarantee external feature transportability.

CKD STAGE RESEARCH STATUS

Binary CKD-like classification: AVAILABLE — RESEARCH ONLY. Stage 1–5 prediction: NOT VALIDATED / NOT AVAILABLE. Stage cannot be inferred from the current Q-CARE binary classifier.

AUTOMATED EVIDENCE SUMMARY

Deterministic wording assembled from current outputs, frozen artifacts, and claim boundaries.

2 OF 3 AGREE. Two of the three research models produced the same class output. Model disagreement should be reviewed alongside model-specific evidence.

QSVC was internally competitive in the frozen benchmark but did not demonstrate a practical quantum advantage. External transportability was not established. The current outputs are binary research classifications only; no CKD stage is produced. Claim registry consistency: PASS.

LIMITATIONS

Small development cohort; development from a single benchmark cohort; missing-heavy clinical dataset; model decision scores are not calibrated disease probabilities; VQC is weak; QSVC has substantial computational cost; external transportability not demonstrated; target comparability with BD-KDD is partial; no Stage 1-5 validation; no clinical deployment validation; research use only.

RESPONSIBLE USE

This report summarizes a Q-CARE research experiment. It is not a medical diagnosis, treatment recommendation, triage decision, validated disease probability, or clinically validated CKD staging result.